

Python Programming — Day 4 Homework

Operators · Comparisons · Logical operators · if / elif / else · API call

Name: _____

Submit before next class

CORE TASKS — COMPLETE ALL 5

1 Arithmetic explorer

Easy

Set **a = 17** and **b = 5**. Print the result of all 7 arithmetic operations with a label for each line.

```
Expected output:
Addition : 22
Subtraction : 12
Multiplication: 85
Division : 3.4
Floor division: 3
Modulus : 2
Power : 1419857
```

Hint: use `print("Addition :", a + b)` style to align labels neatly.

2 Odd or even checker

Easy

Take a number as input from the user. Use the modulus operator to check if it is odd or even and print the result.

```
Enter a number: 7
7 is odd
```

```
Enter a number: 12
12 is even
```

Hint: a number is even if `number % 2 == 0`. Remainder 0 means it divides exactly.

3 Grade calculator

Medium

Take a student's marks (0–100) as input. Print the grade using if/elif/else. Print "Invalid marks" if outside 0–100.

```
90 - 100 → Grade: A
75 - 89 → Grade: B
60 - 74 → Grade: C
45 - 59 → Grade: D
Below 45 → Grade: F
```

Hint: `input()` always returns a string — convert to `int` first.

4 Login system

Medium

Extend the class login example by adding a third field — a secret code. Print a specific message for each case.

```
username = "admin"
password = "pass123"
secret_code = "7777"
```

```
All 3 correct → Welcome! Login successful.
Username wrong → Username not found.
Password wrong → Incorrect password.
Secret wrong → Wrong secret code.
All 3 wrong → Access denied.
```

Hint: 3 variables = 8 combinations (2^3). Use `elif` and think about order of checks.

5 GitHub API — conditional profile card

Challenge

Make the GitHub API call. Then use if/else to print a personalised summary based on the live data you receive.

```
import requests

url = "https://api.github.com/users/YOUR_USERNAME"
data = requests.get(url).json()

# Apply these conditions:
# 1. followers > 10 → "Popular user!" else → "Keep growing!"
# 2. public_repos > 5 → "Active contributor!" else → "Just getting started!"
# 3. location is not None → print it else → "Location not set"
# 4. Print name, login, and followers count in a clean format
```

Hint: `data.get("location")` returns `None` if empty. Check with `if data.get("location") is not None`. Use `data.get("followers", 0)` to set a default of 0 if key is missing.

BONUS — OPTIONAL CHALLENGE

B Simple ATM simulator

Optional

Set a fixed PIN (e.g. **1234**) and a balance (e.g. **5000**). Ask the user for their PIN. If correct, ask how much to withdraw. Use conditions to check: is the PIN right? Is the amount positive? Is there enough balance? Print the updated balance or an appropriate error message.